MICROECONOMICS · INDUSTRIAL ORGANIZATION · LECTURE

Oligopoly & the Logic of the Future

Cournot, Bertrand, and Repeated Games

- Monopoly
- Cournot
- Competition / Bertrand

ROADMAP

Where we are going

PART A

Cournot duopoly

Firms choose quantities. Best responses, the symmetric NE, N firms, Stackelberg.

PART B

Bertrand duopoly

Firms choose prices. The paradox, and three ways out: capacity, differentiation, repetition.

PART C

Sequential & finite repetition

Game trees, backward induction, SPE — and why a finite horizon unravels.

PART D

Infinite repetition

Discounting, trigger strategies, the threshold δ^*, and the folk theorem.

REFERENCE

Key results, at a glance

COURNOT (symmetric)

$q^* = \dfrac{a-c}{3b} \quad P^* = \dfrac{a+2c}{3}$

$\pi^* = \dfrac{(a-c)^2}{9b} \quad N \text{ firms: } q = \dfrac{a-c}{(N+1)b}$

BERTRAND

Homogeneous: $p^* = c, \ \pi = 0$

Differentiated: $p^* = \dfrac{a+bc}{2b-d} > c$

REPEATED PD

Finite T, unique stage NE $\Rightarrow$ defect always

Infinite: cooperate iff $\delta \geq \delta^* = \dfrac{T-R}{T-P}$

MONOPOLY / CARTEL

$Q^m = \dfrac{a-c}{2b} \quad P^m = \dfrac{a+c}{2}$

$\pi^m = \dfrac{(a-c)^2}{4b}$ — not a Nash equilibrium

STACKELBERG

$q_L = \dfrac{a-c}{2b} > q_F = \dfrac{a-c}{4b}$

$\pi_L = \dfrac{(a-c)^2}{8b} > \pi_F = \dfrac{(a-c)^2}{16b}$

THE ORDERING

$P^{\text{Bertrand}} < P^{\text{Cournot}} < P^{\text{Monopoly}}$

the strategic variable matters

PART A

Cournot Duopoly

Quantity competition

Firms simultaneously choose how much to produce; the market clears at whatever price total output commands. Output lands between monopoly and competition.

COURNOT · SETUP

Inverse demand & constant costs

- Two firms make a **homogeneous good**; each picks a quantity $q_i \geq 0$ simultaneously.
- Price is determined by **total** output through the inverse demand curve.
- Marginal cost is constant and identical, with $0 \leq c < a$ and $b > 0$; no fixed costs.
- Each firm maximizes profit taking the rival's quantity **as given** — a Nash mindset.

INVERSE DEMAND

$$P = a - b\,(q_1 + q_2)$$

COST & PROFIT

$$\pi_i = \big[a - b(q_i + q_j) - c\big]\, q_i$$

Profit, the FOC, and marginal revenue

- Differentiate profit in q_i: $\dfrac{\partial \pi_i}{\partial q_i} = a - 2bq_i - bq_j - c = 0.$
- Second-order condition $\dfrac{\partial^2 \pi_i}{\partial q_i^2} = -2b < 0$ — strictly concave, so the FOC pins a maximum.
- Precision point: marginal revenue is the price on the last unit minus the price hit on the units already being sold.

MARGINAL REVENUE, DECOMPOSED

$$MR_i = \underbrace{(a - bq_j - bq_i)}_{\text{price } P \text{ on marginal unit}} + \underbrace{(-bq_i)}_{\text{loss on inframarginal units}}$$
$$= a - bq_j - 2bq_i$$

The reaction function

- Solve the FOC for q_i to get the **best response** to any q_j.
- The slope is $-\frac{1}{2}$: each unit the rival adds, I withdraw half a unit.
- Negative slope $\Rightarrow$ quantities are **strategic substitutes.**
- The intercept $\frac{a-c}{2b}$ is exactly the monopoly output — what I'd make if the rival vanished.

BEST RESPONSE

$$q_i = R_i(q_j) = \frac{a-c}{2b} - \frac{1}{2}\,q_j$$

COURNOT · EQUILIBRIUM

The symmetric Nash equilibrium

Set $q_1 = q_2 = q^*$ in the best response:

$$q^* = \frac{a-c}{2b} - \frac{1}{2}q^*.$$

Collecting terms, $\frac{3}{2}q^* = \dfrac{a-c}{2b}$, so each firm produces

$$q^* = \frac{a-c}{3b}.$$

Neither firm can profitably deviate — each is on its own reaction curve.

EQUILIBRIUM VALUES

$$Q^* = \frac{2(a-c)}{3b} \qquad P^* = \frac{a+2c}{3}$$

$$\pi^* = \frac{(a-c)^2}{9b} \text{ per firm}$$

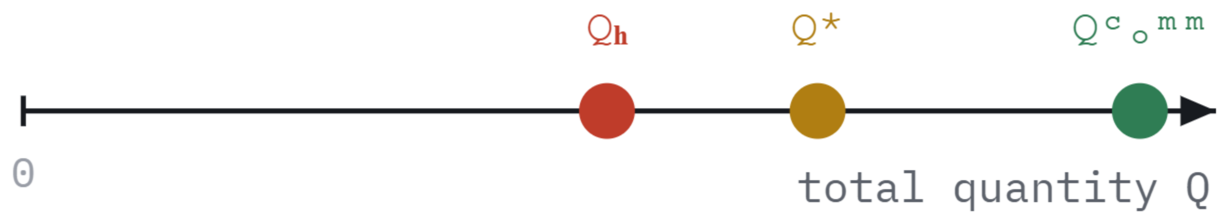
COURNOT · BENCHMARKS

Three benchmarks, one ordering

MONOPOLY $Q^m = \dfrac{a-c}{2b}, \quad P^m = \dfrac{a+c}{2}, \quad \pi^m = \dfrac{(a-c)^2}{4b}$

COURNOT $Q^* = \dfrac{2(a-c)}{3b}, \quad P^* = \dfrac{a+2c}{3}, \quad \pi^* = \dfrac{(a-c)^2}{9b}$

COMPETITION $P = c, \quad Q = \dfrac{a-c}{b}, \quad \pi = 0$



$$Q^m < Q^{\text{Cournot}} < Q^{\text{comp}}$$

The cartel is not a Nash equilibrium

- Split the cartel evenly: each makes $\frac{Q^m}{2} = \dfrac{a-c}{4b}$.
- But my **best response** to $q_j = \frac{a-c}{4b}$ is to produce **more**, not the cartel share.
- Each firm has a private incentive to **secretly expand** — the cartel unravels.
- This is why sustained collusion needs **repetition** (Part D), not goodwill.

DEVIATION FROM THE CARTEL SPLIT

$$R_i\left(\tfrac{a-c}{4b}\right) = \frac{a-c}{2b} - \frac{a-c}{8b} = \frac{3(a-c)}{8b}$$

$$\frac{3(a-c)}{8b} > \frac{a-c}{4b} \quad \text{\small(expand!)}$$

Cost asymmetry & exit

- With distinct costs $c_i \neq c_j$, the FOCs give the **asymmetric equilibrium** at right.
- The **lower-cost** firm produces more and earns more — efficiency is rewarded.
- If $c_i \geq \frac{a+c_j}{2}$ the formula turns negative: firm i **exits** ($q_i^* = 0$).
- Then j is a **monopolist** at $\frac{a-c_j}{2b}$ — output jumps discontinuously.

ASYMMETRIC EQUILIBRIUM

$$q_i^* = \frac{a - 2c_i + c_j}{3b}$$

EXIT CONDITION

$$c_i \geq \frac{a + c_j}{2} \;\Rightarrow\; q_i^* = 0$$

COURNOT · WELFARE

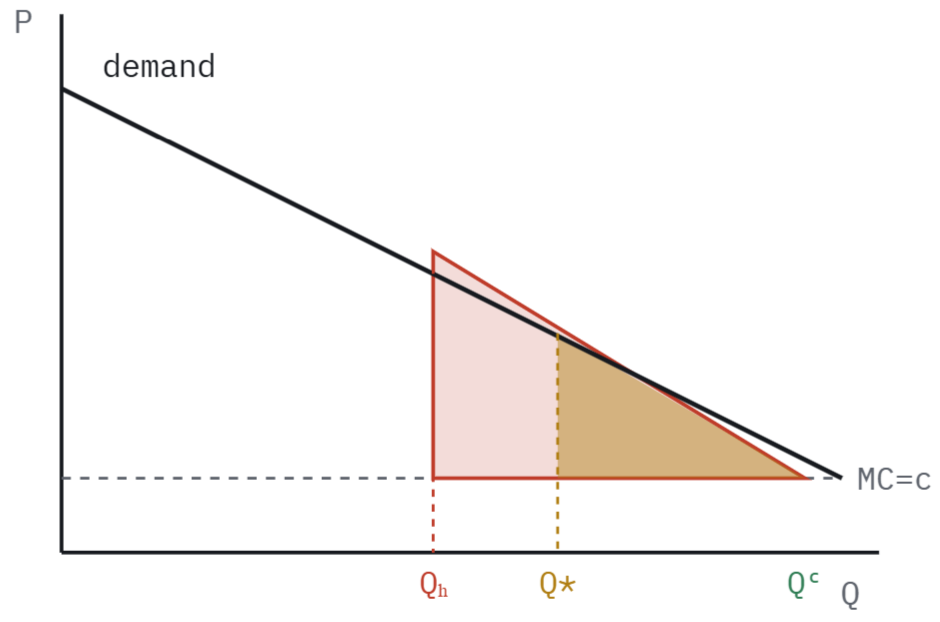
Deadweight loss

The lost surplus is the triangle between price and marginal cost over the missing units:

$$DWL = \frac{1}{2}\left(Q^{\text{comp}} - Q\right)\left(P - c\right)$$

COURNOT	MONOPOLY
$\dfrac{(a-c)^2}{18b}$	$\dfrac{(a-c)^2}{8b}$

Cournot loss $<$ monopoly loss: more output, less waste.



N firms: Cournot $\to$ competition

- Symmetric N-firm equilibrium output: $q^* = \dfrac{a-c}{(N+1)b}$.
- Price $P^* = \dfrac{a+Nc}{N+1}$ and markup $\dfrac{a-c}{N+1} \to 0$ as $N \to \infty$.
- $N = 1$ recovers monopoly; $N \to \infty$ recovers perfect competition.
- Market power, measured by the Lerner index, falls with entry.

LERNER INDEX

$$L = \frac{P-c}{P} = \frac{a-c}{a+Nc} \xrightarrow[N\to\infty]{} 0$$

Stackelberg: the value of commitment

- Leader picks q_L; follower observes and plays $q_F = R(q_L)$.
- **Backward induction**: the leader maximizes profit **substituting in** the follower's reaction.
- The leader earns **twice** the follower's profit — a first-mover advantage.
- The edge is **commitment**: the leader cannot be undercut on a choice already made.

QUANTITIES

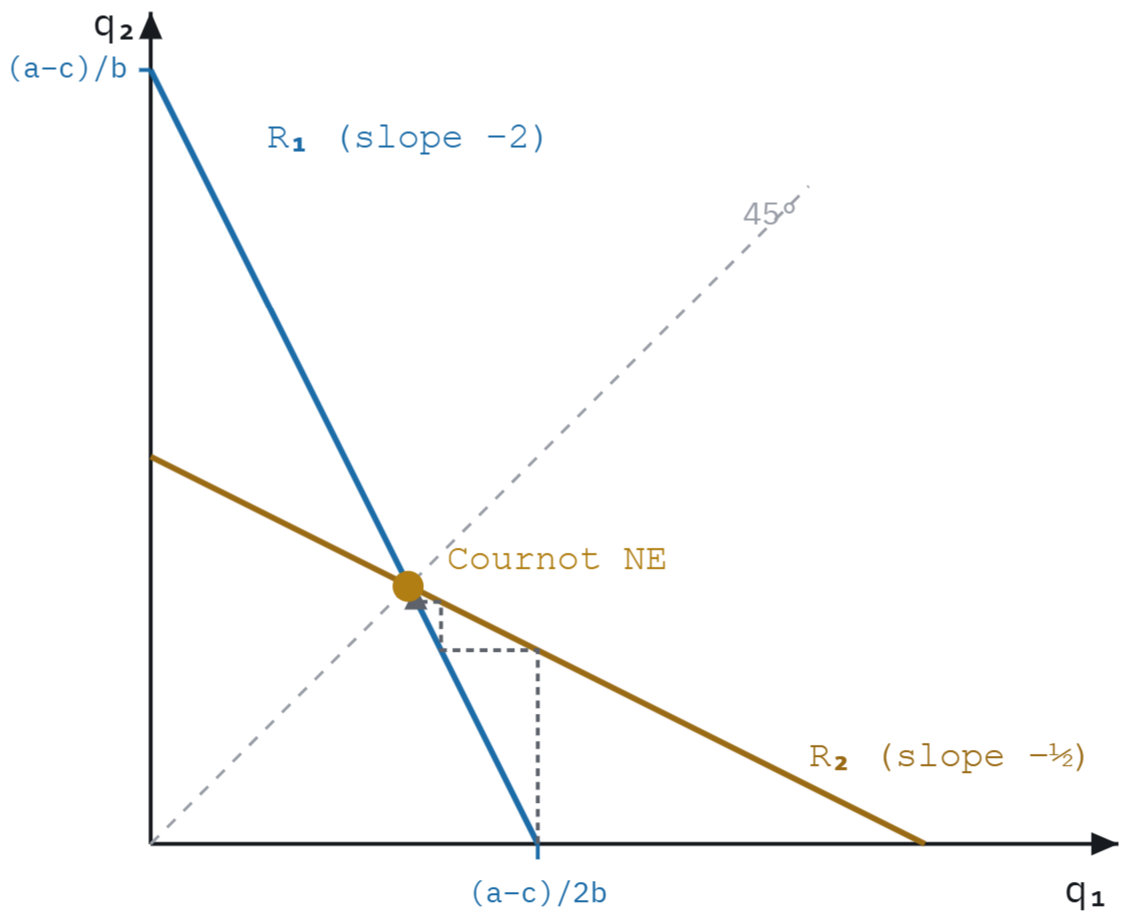
$$q_L = \frac{a-c}{2b} > q_F = \frac{a-c}{4b}$$

PROFITS

$$\pi_L = \frac{(a-c)^2}{8b} > \pi_F = \frac{(a-c)^2}{16b}$$

The reaction-function diagram

- R_1: slope -2, the steeper curve, from $\left(\frac{a-c}{2b}, 0\right)$ to $\left(0, \frac{a-c}{b}\right)$.
- R_2: slope $-\frac{1}{2}$, the flatter curve.
- They cross once on the 45° line — the symmetric NE $\left(\frac{a-c}{3b}, \frac{a-c}{3b}\right)$.
- Tâtonnement spirals inward (stable, since $\frac{1}{2} \cdot \frac{1}{2} < 1$) — dynamics, not the equilibrium concept.



COURNOT · WORKED EXAMPLE

Numbers: $a = 120,\ b = 1,\ c = 0$

	Monopoly	Cournot	Competition	Stackelberg
Total Q	60	80	120	90
Price P	60	40	0	30
Profit	3600	1600 each	0	1800 / 900
DWL	1800	800	0	450

▸ Check the Cournot column

PUZZLE 1 · CLICKER

The Traveler's Dilemma

Two travelers lose identical antiques. Each independently writes a claim, an integer from 2 to 100. Both are reimbursed the lower claim; the low claimant gets a $+2$ bonus, the high claimant a -2 penalty (ties: both get the claim).

What is the Nash equilibrium? What do real people play?

▸ Reveal solution

PUZZLE 2 · CLICKER · RUN LIVE

The p-Beauty Contest

Everyone in the room picks a number in $[0, 100]$. The winner is whoever lands closest to $\frac{2}{3}$ of the average of all guesses.

What number do you choose — and what is the equilibrium?

▸ Reveal solution

PART B

Bertrand Duopoly

Price competition

When firms compete in prices, the temptation to undercut is overwhelming — and it drives price all the way to marginal cost with only two sellers.

Prices, undercutting & the demand rule

- Homogeneous good; firms simultaneously post prices p_1, p_2.
- Consumers buy entirely from the cheaper firm; a tie splits demand 50/50.
- Profit is $\pi_i = (p_i - c)\, D_i(p_i, p_j)$.
- A firm can seize the whole market by undercutting the rival by a penny.

DEMAND ALLOCATION

$$D_i = \begin{cases} D(p_i) & p_i < p_j \\ \frac{1}{2} D(p_i) & p_i = p_j \\ 0 & p_i > p_j \end{cases}$$

The Bertrand paradox

- With $c_1 = c_2 = c$, the unique NE is $p_1 = p_2 = c$ and zero profit.
- Just two firms reproduce the perfectly competitive outcome.
- Price is independent of n for any $n \geq 2$ — more entrants don't help consumers further.
- Compare Cournot, where price stays above cost for any finite N.

UNIQUE EQUILIBRIUM

$$p_1^* = p_2^* = c, \qquad \pi = 0$$

BERTRAND · MECHANISM

Why? Discontinuous demand

- No interior FOC: a firm's payoff jumps as its price crosses the rival's.
- At any $p > c$, undercut by ε to capture the whole market — strictly profitable.
- At $p = c$, undercutting means a loss and raising means zero sales — no deviation pays.
- The 50/50 tie rule is what makes (c, c) a clean equilibrium.

DEVIATION INCENTIVE AT $P > C$

$$\pi_i(p-\varepsilon) \approx (p-c)\,D(p)$$
$$\gg \tfrac{1}{2}(p-c)\,D(p) = \pi_i(p)$$

Asymmetric costs break zero profit

- Let $c_1 < c_2$. The low-cost firm prices **just below** c_2.
- It captures the **entire market** and earns a positive margin $\approx c_2 - c_1$.
- The inefficient firm is priced out — it cannot profitably undercut.
- So "$P = MC$, zero profit" is **special to identical costs**.

LOW-COST FIRM

$$p_1 \to c_2^-, \qquad \pi_1 \approx (c_2 - c_1)\, D(c_2) > 0$$

Capacity constraints

- Firms pick capacities k_i first, then prices — a two-stage game.
- The higher-priced firm serves only residual demand.
- Kreps–Scheinkman: under efficient rationing, the outcome equals Cournot.
- With tight capacities, no pure-strategy NE exists $\Rightarrow$ Edgeworth price cycles.

RESIDUAL DEMAND

$$D_i^{\text{res}} = \max\{\, D(p_i) - k_j,\ 0\,\}$$

Product differentiation

Demand $q_i = a - b\,p_i + d\,p_j$ with $0 < d < 2b$.

FOC: $a - 2b\,p_i + d\,p_j + bc = 0$, so

$$p_i = \frac{a + bc + d\,p_j}{2b} \qquad \text{(rising in } p_j\text{)}$$

Symmetry $\Rightarrow$ $p^* = \dfrac{a+bc}{2b-d} > c,\ \pi^* > 0.$

WORKED: $A = 100,\ B = 1,\ D = 0.5,\ C = 20$

$$p^* = \frac{100 + 20}{1.5} = 80$$

$$q^* = 60, \qquad \pi^* = (80 - 20) \cdot 60 = 3600$$

BERTRAND · PRECISION POINT

Differentiated prices are strategic complements

- Best responses slope up: a higher rival price lets me raise mine — strategic complements.
- Opposite of Cournot, where quantities are substitutes (downward-sloping).
- Do not say $p^* \to c$ as $d \to 2b$ — here p^* rises in d.
- The homogeneous paradox is a separate limiting case, not this limit.

COMPARATIVE STATIC

$$\frac{\partial p^*}{\partial d} = \frac{a + bc}{(2b - d)^2} > 0$$

Repeated interaction sustains collusion

- Grim trigger: cooperate at p^m until someone undercuts, then revert to $p = c$ forever.
- Collusion is sustainable when firms value the future enough.
- Duopoly threshold: $\delta \geq \frac{1}{2}$. For n firms: $\delta \geq 1 - \frac{1}{n}$.
- More firms $\Rightarrow$ harder to collude. Full theory in Part D.

SUSTAINING P^M

$$\frac{\pi^m/2}{1-\delta} \geq \pi^m \;\Rightarrow\; \delta \geq \tfrac{1}{2}$$

$$n \text{ firms: } \delta \geq 1 - \tfrac{1}{n}$$

SYNTHESIS

Cournot vs Bertrand, head to head

Homogeneous good · $P = A - Q$ · marginal cost c

	Bertrand	Cournot	Monopoly
Price	c	$c + \dfrac{A-c}{3}$	$\dfrac{A+c}{2}$
Profit / firm	0	$\dfrac{(A-c)^2}{9}$	$\dfrac{(A-c)^2}{4}$

$$P^B < P^C < P^M$$

The strategic variable matters as much as firm count: undercutting (discontinuous demand) bites harder than quantity expansion.

PUZZLE 3 · CLICKER

Which model gives the lowest price?

Take $a = 100,\ b = 1,\ c = 10$. Rank the equilibrium prices under monopoly, Cournot, and Bertrand.

Put them in order — and by how much do they differ?

▸ Reveal solution